

MATH 121: COMMUTATIVE ALGEBRA

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1. RINGS

Important Example: Let X be a “geometric set”, and consider the ring

$$\text{Hom}(X, \mathbb{R}) = \{X \xrightarrow{f} \mathbb{R}\}$$

with pointwise addition and multiplication

$$(f + g)(x) = f(x) + g(x), \quad (fg)(x) = f(x)g(x)$$

$r \in \text{Hom}(X, \mathbb{R})$ is a unit if and only if there exists $g \in \text{Hom}(X, \mathbb{R})$ such that $fg = 1$ if and only if $\forall x \in X, (fg)(x) = f(x)g(x) = 1$

$$\text{Hom}(X, \mathbb{R})^\times = \text{Hom}(X, \mathbb{R}^\times)$$

Definition 1.1. Let $a \in R, a \neq 0$

- a is a unit if and only if $\exists b \in R$ such that $ab = 1$,
- a is a zero divisor if and only if $\exists b \in R \setminus \{0\}$ such that $ab = 0$,

Lemma 1.2. The set of units $R^\times = \{a \in R \mid a \text{ is a unit}\}$ is a group under multiplication.

Proof: Worksheet (03/31).

Lemma 1.3 (Useful Lemma). If $\phi : R \rightarrow S$ is a ring homomorphism, then it follows from the definition that

- $\phi(0_R) = 0_S$
- $\phi(-a) = -\phi(a)$
- $\phi(a^{-1}) = \phi(a)^{-1}$

Remark 1.4. $\mathbb{Z}$ is initial in the category of rings. There is a unique homomorphism from the initial object to every other object. For any ring R ,

$$\text{Hom}(\mathbb{Z}, R) = \{1 \rightarrow 1_R\}$$

We don't know anything about homomorphisms to the initial object.

2. IDEALS

Lemma 2.1. If $\phi : R \rightarrow S$ is any ring homomorphism and $J \subset S$ is an ideal $\phi^{-1}(J)$ is also an ideal.

Proof: Worksheet Q4.1.

Theorem 2.2. If $I \subset R$ is an ideal then,

- (1) There exists a ring R/I and a surjective ring homomorphism $\pi : R \rightarrow R/I$ such that $\ker(\pi) = I$. Moreover R/I and π are unique upto isomorphism.
- (2) There is an inclusion preserving bijection

$$\{\text{Ideals in } R/I\} \leftrightarrow \{\text{Ideals in } R \text{ containing } I\}$$

$$\bar{J} \leftrightarrow \pi^{-1}(\bar{J})$$

Proof.

- (1) Construct R/I explicitly.
- (2) I gives an equivalence relation $\sim$ on R such that

$$a \sim b \iff a - b \in I \iff a = b + I$$

Show that equivalence classes $R/\sim$ is a ring.

$$R \rightarrow R/I = R/\sim$$

$$a \mapsto \bar{a} = a + I$$

Let $\bar{J}$ be an ideal in R/I , and J be an ideal in R and define the maps

$$\phi : \bar{J} \rightarrow \pi^{-1}(\bar{J}), \quad \psi : J \rightarrow \pi(J)$$

Now we want to show that $\phi \circ \psi = \psi \circ \phi = \text{id}$. Consider $\phi \circ \psi(J)$ and $\psi \circ \phi(\bar{J})$. Note that

$$\psi \circ \phi(\bar{J}) = \pi(\pi^{-1}(\bar{J})) = \bar{J}$$

since π is surjective. Thus $\psi \circ \phi = \text{id}$.

Now, note that $J \subseteq \phi \circ \psi(J) = \pi^{-1}(\pi(J))$, and since $\bar{0} \in \pi(J)$, $I \subseteq \pi^{-1}(\pi(J))$. An element $x \in \pi^{-1}(\bar{J})$ looks like $x + I$, and $\pi^{-1}(\bar{J}) = J + I$. Moreover since $\bar{0} \in \bar{J}$, we have $\pi^{-1}(\bar{0}) \subseteq \pi^{-1}(\bar{J})$ which implies $I \subseteq J$. Thus, $\psi \circ \phi(J) = J$ and $\psi \circ \phi = \text{id}$.

□

Exercise: Describe the ideals in

- (1) $\mathbb{Z}/\langle 24 \rangle = \{\langle 0 \rangle, \langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 4 \rangle, \langle 6 \rangle, \langle 8 \rangle, \langle 12 \rangle\}$
- (2) $\mathbb{Z}/\langle n \rangle = \{\langle d \rangle : d|n\}$
- (3) $\mathbb{Z}/\langle 3 \rangle = \{\langle 0 \rangle, \langle 1 \rangle\}$
- (4) $\mathbb{Z}/\langle 5 \rangle = \{\langle 0 \rangle, \langle 1 \rangle\}$
- (5) $\mathbb{C}[x, y]/\langle y - x^2 \rangle \cong \mathbb{C}[x] = \{\text{Ideals in } \mathbb{C}[x]\}$
- (6) $\mathbb{C}[x]/\langle x^n \rangle = \{\langle x^i \rangle : 0 \leq i < n\}$

Lemma 2.3. If $\phi : R \rightarrow S$ is a ring homomorphism with $\ker(\phi) \supseteq I$ then there exists a unique morphism $\tilde{\phi} : R/I \rightarrow S$

$$\begin{array}{ccc} R & \xrightarrow{\pi} & R/I \\ \phi \downarrow & \swarrow \tilde{\phi} & \\ S & & \end{array}$$

Universal Property of Quotients: Upto isomorphism R/I is the ring such that

$$\text{Hom}(R/I, S) = \{\phi : R \rightarrow S : I \subset \ker(\phi)\}$$

We want to define $\tilde{\phi}(\bar{a}) = \tilde{\phi}(a + I) = \phi(a)$ and $\tilde{\phi}(\bar{0}) = 0$.

Lemmat 2.4. Maximal Ideal $\implies$ Prime Ideals.

Definition 2.5. The spectrum of a ring R is the set

$$\text{Spec}(R) = \{\text{Prime Ideals in } R\}$$

the maximal spectrum of R is the set

$$\mathfrak{m}\text{Spec}(R) = \{\text{Maximal Ideals in } R\}$$

$\text{Spec}(R)$ is where the ‘geometry lives’. Note that $\mathfrak{m}\text{Spec}(R) \subset \text{Spec}(R)$.

Example: $R = \mathbb{Z}[\sqrt{-5}]$. 2 is irreducible but not prime. $R/\langle 2 \rangle = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Example: $R = \mathbb{C}[x, y]/\langle x^3 - y^2 \rangle$. x is irreducible, but $\langle x \rangle$ is not prime. $R/\langle x \rangle = \mathbb{C}[y]/\langle y \rangle$ and y is a zero divisor.

Theorem 2.6. Let R be a non-zero ring.

- (1) R has a maximal ideal.
- (2) If $I \subsetneq R$ is a proper ideal there exists a prime ideal $P \subseteq R$ such that $I \subset P \subsetneq R$.

Proof. **Zorn’s Lemma:** Let $(P, \leq)$ be a po-set.

If $P \neq \emptyset$ and if $C = C_1 \leq C_2 \leq C_3 \leq \dots$ is a chain in $(P, \leq)$ then there exists $B \in P$ such that $C_i \leq B$ for all i , then there exists $M \in P$ such that if $S \leq M$ then $M = S$.

Let $P = \{\text{Proper ideals in } R\}$ and $\leq$ be inclusion. We claim that $(P, \leq)$ is a poset.

- (1)
 - $P \neq \emptyset$ since $\langle 0 \rangle \in P$.
 - Given a chain in P

$$I_1 \subseteq I_2 \subseteq \dots$$

Then, define $B = \bigcup I_j$. Note that B is an ideals since union of nested ideals is an ideal. And B is proper since $1 \notin I_j$ for all j , and hence $1 \notin B$. Then there exists some ideal $M \subset R$ such that M is maximal.

- (2) Follows from (1). Let $R = R/I$. Pre-image of prime ideals is prime.

Thus, $\mathfrak{m}\text{Spec}(R) \neq \emptyset$ and $\text{Spec}(R) \neq \emptyset$. □

3. ZARISKI TOPOLOGY

Definition 3.1. If $S \subseteq R$ is any subset of R ,

$$\mathbb{V}(S) = \{[P] \in \text{Spec}(R) | S \subseteq P\} \subseteq \text{Spec}(R)$$

Definition 3.2. An algebraic subset of $\mathbb{K}^n$ of some field $\mathbb{K}$ is a set of the form

$$\mathbb{V}(S) = \{P \in \mathbb{K}^n | f(p) = 0, \forall f \in S\}$$

for a subset $S \subset \mathbb{K}[x_1, \dots, x_n]$

Examples:

- $\mathbb{V}(x^2 + y^2 - 1) \subseteq \mathbb{R}^2$

$$\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 - 1 = 0\}$$

- $\mathbb{V}(x^2 + y^2 - 1) \subseteq \mathbb{R}^3$

$$\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 - 1 = 0\}$$

- $\mathbb{V}(x^2 - y) \subseteq \mathbb{R}[x, y]$

$$\{(x, y) \in \mathbb{R}[x, y] \mid x^2 = y\}$$

- $\mathbb{V}(x^2 - xy) \subseteq \mathbb{R}[x, y]$

$$\{(x, y) \in \mathbb{R}[x, y] \mid x = 0, x - y = 0\}$$

- $\mathbb{V}(x^2 + 1) \subseteq \mathbb{R}[x]$

$$\mathbb{V}(x^2 + 1) = \emptyset$$

- $\mathbb{V}(x^2 + 1) \subseteq \mathbb{C}[x]$

$$\mathbb{V}(x^2 + 1) = \{i, -i\}$$

- $\mathbb{V}(x^2 + 4) \subseteq \mathbb{F}_3[x]$

$$\mathbb{V}(x^2 + 4) = \emptyset$$

- $\mathbb{V}(x^2 + 4) \subseteq \mathbb{F}_p[x]$

TODO:

- $\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) \subseteq \mathbb{R}[x, y]$

$$\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) = \mathbb{V}(x^3(x - y)^3) = \{(x, y) \in \mathbb{R}[x, y] \mid x = 0, x - y = 0\}$$

- $\mathbb{V}(n(x^2 + y^2 + z^2), n \in \mathbb{Z}) \subseteq \mathbb{R}^3$

TODO:

- $\mathbb{V}(\text{a collection of any linear polynomials in } \mathbb{K}[x_1, \dots, x_n])$

TODO:

Definition 3.3. Let $V \subseteq \mathbb{K}^n$ be an algebraic set. The defining ideal of V is

$$\mathbb{I}(V) = \{f \in \mathbb{K}[x_1, \dots, x_n] \mid f(p) = 0, \forall p \in V\}$$

Definition 3.4. If $\mathfrak{M} \subseteq R$ is a maximial ideal, the residue field of $\mathfrak{M}$ is $R/\mathfrak{M}$.

Definition 3.5. Let $\mathbb{K}$ be a field. A $\mathbb{K}$ algebra is the data of a ring R together with a fixed ring homomorphism

$$\alpha : \mathbb{K} \rightarrow R$$

Definition 3.6. If $(R, \mathbb{K} \xrightarrow{\alpha} R)$ and $(S, \mathbb{K} \xrightarrow{\beta} S)$ are $\mathbb{K}$ -algebras, a ring map $\phi : R \rightarrow S$ is a morphism of $\mathbb{K}$ -algebras iff

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{\phi} & \mathbb{S} \\ \alpha \uparrow & \nearrow \beta & \\ \mathbb{K} & & \end{array}$$

Lemmact 3.7. If $\mathbb{R}$ is $\mathbb{k}$ -algebra, $\mathbb{R}/I$ has a natural $\mathbb{k}$ -algebra structure compatible with the quotient map.

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{\pi} & \mathbb{R}/I \\ \alpha \uparrow & \nearrow \exists! & \\ \mathbb{k} & & \end{array}$$

Definition 3.8. A valuation ring of a field $\mathbb{K}$ is a subring $\mathbb{R} \subset \mathbb{K}$ with the property that for all $x \in \mathbb{K}^\times$ either $x \in \mathbb{R}$ or $x^{-1} \in \mathbb{R}$.

4. GRÖBNER BASIS

Remark 4.1. *Ideal Membership Problem:* Suppose $I \subseteq \mathbb{R}[x_1, \dots, x_n]$ an ideal. How can we algorithmically determine if $f \in I$?

Definition 4.2. A ring $\mathbb{R}$ is Noetherian if and only if every ascending chain of ideals stabilizes (Ascending Chain Condition), i.e.

$$I_1 \subseteq I_2 \subseteq \dots$$

there exists $N \in \mathbb{Z}$ such that $I_k = I_{k+1}$ for all $k \geq N$.

Lemma 4.3. $\mathbb{R}$ is Noetherian if and only if every ideal is finitely generated.

Lemma 4.4 (Dickson's Lemma). Every monomial ideal in $\mathbb{K}[x_1, \dots, x_n]$ is finitely generated.

Theorem 4.5 (Hilbert's Basis Theorem). If $\mathbb{R}$ is a Noetherian ring, then $\mathbb{R}[x]$ is also Noetherian.

5. NULLSTELLENSATZ

Theorem 5.1 (Hilbert's Nullstellensatz I). If $\mathbb{K}$ is algebraically closed, then every maximal ideal of $\mathbb{K}[x_1, \dots, x_n]$ is of the form $\langle x_1 - a_1, \dots, x_n - a_n \rangle$ for some point $(a_1, \dots, a_n) \in \mathbb{K}^n$.

Theorem 5.2 (Hilbert's Nullstellensatz II). If $\mathbb{K}$ is algebraically closed and $V = V(\mathcal{S}) \subset \mathbb{K}^n$ is an algebraic set, for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, then $I(V) = \sqrt{\langle \mathcal{S} \rangle}$.

Theorem 5.3 (Hilbert's Weak Nullstellensatz). If $\mathbb{K}$ is algebraically closed and $I \subset \mathbb{K}[x_1, \dots, x_n]$ is an ideal then $V(I) = \emptyset$ if and only if $I = \langle 1 \rangle$.

Theorem 5.4 (Hilbert's Nullstellensatz III). Let $\mathbb{K}$ be an algebraically closed field. If $V \subset \mathbb{K}^n$ is an algebraic subset then the maps I_V and V_V induce inverse, inclusion-reversing bijections:

$$\begin{array}{ccc} \left\{ \begin{array}{c} \text{algebraic subsets} \\ \text{in } V \end{array} \right\} & \begin{array}{c} \xrightarrow{I_V} \\ \xleftarrow{V_V} \end{array} & \left\{ \begin{array}{c} \text{radical ideals} \\ \text{in } \mathbb{K}[V] \end{array} \right\} \\ & 5 & \end{array}$$

Remark 5.5. $\mathbb{I}$ and $\mathbb{V}$ properties:

(1) If $\mathcal{S} \subset \mathcal{T}$ then $\mathbb{V}(\mathcal{T}) \subset \mathbb{V}(\mathcal{S})$

(2)

$$\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle) = \mathbb{V}(\sqrt{\langle \mathcal{S} \rangle})$$

(3) $\mathbb{I}(\mathbb{V})$ is radical.

(4) $\langle \mathcal{S} \rangle \subset \sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(\mathbb{V}(\mathcal{S}))$

(5) $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\mathbb{I}(\mathbb{V}(\mathcal{S})))$

(6) If $fg \in \mathbb{I}(\mathbb{V})$ then $\mathbb{V} \subseteq \mathbb{V}(g) \cap \mathbb{V}(f)$.

6. LOCALIZATION

Definition 6.1. A subset $\mathcal{U} \subset \mathcal{R}$ is **multiplicatively closed** if and only if

(1) $1_{\mathcal{R}} \in \mathcal{U}$

(2) and if $s, u \in \mathcal{U}$ then $su \in \mathcal{U}$.

Example 6.1.

(1) $\mathcal{R} = \mathbb{Z}$, $\mathcal{U} = \{1, 3, 9, \dots\}$ is multiplicatively closed.

(2) $\mathcal{R} = \mathbb{Z}$, $\mathcal{U} = \{1, 3, 9\}$ is not multiplicatively closed.

(3) $\mathcal{R} = \mathbb{R}[x, y]$, $\mathcal{U} = \{f \in \mathcal{R} \mid f(1, 0) \neq 0\}$ is multiplicatively closed.

(4) $\mathcal{R} = \mathbb{R}[x, y]$, $\mathcal{U} = \{f \in \mathcal{R} \mid f(1, 1) \neq 0, \text{ or } f(0, 0) \neq 0\}$ is not multiplicatively closed.

Definition 6.2. Let $\mathcal{R}$ be a ring and $\mathcal{U} \subseteq \mathcal{R}$ be a multiplicative closed set. The **localization** of $\mathcal{R}$ at $\mathcal{U}$ is a ring $\mathcal{U}^{-1}\mathcal{R}$ together with a ring map $\eta : \mathcal{R} \rightarrow \mathcal{U}^{-1}\mathcal{R}$ that is universal in the following sense:

(1) Given a ring map $\psi : \mathcal{R} \rightarrow \mathcal{T}$ such that $\psi(\mathcal{U}) \subseteq \mathcal{T}^\times$ there exists a unique ring map $\tilde{\psi} : \mathcal{U}^{-1}\mathcal{R} \rightarrow \mathcal{T}$ such that

$$\begin{array}{ccc} \mathcal{R} & \xrightarrow{\eta} & \mathcal{U}^{-1}\mathcal{R} \\ & \searrow \psi & \downarrow \tilde{\psi} \\ & & \mathcal{T} \end{array}$$

Construction of $\mathcal{U}^{-1}\mathcal{R}$:

Consider $\mathcal{R} \times \mathcal{U}$ as a set (r_1, u_1) where $r_1 \in \mathcal{R}, u_1 \in \mathcal{U}$ “=” “ $\frac{r_1}{u_1}$ ”. Define an equivalence relation

$$(r_1, u_1) \sim (r_2, u_2) \iff t(u_2 r_1 - u_1 r_2) = 0, \quad t \in \mathcal{U}$$

Note if $\mathcal{R}$ is a domain $t = 1_{\mathcal{R}}$.

Define $\frac{r}{u}$ to be the equivalence class of (r, u) in $\mathcal{R} \times \mathcal{U} / \sim$.

Set $\mathcal{U}^{-1}\mathcal{R} = \mathcal{R} \times \mathcal{U} / \sim = \left\{ \frac{r}{u} \mid r \in \mathcal{R}, u \in \mathcal{U} \right\} / \sim$

Make $\mathcal{U}^{-1}\mathcal{R}$ a ring with

$$\left(\frac{r_1}{u_1} \right) \left(\frac{r_2}{u_2} \right) = \frac{r_1 r_2}{u_1 u_2}, \quad \frac{r_1}{u_1} + \frac{r_2}{u_2} = \frac{r_1 u_2 + r_2 u_1}{u_1 u_2}$$

Additive Identity: 0/1, **Multiplicative Identity:** 1/1

$$\eta : R \rightarrow U^{-1}R$$

Then

$$\begin{aligned} \ker(\eta) &= \{r \in R \mid \eta(r) = 0\} \\ &= \{r \in R \mid \frac{r}{1} = \frac{0}{1}\} \\ &= \{r \in R \mid t(r \cdot 1 - 1 \cdot 0) = 0, t \in U\} \\ &= \{r \in R \mid tr = 0, t \in U\} \end{aligned}$$

If R is a domain, $\ker(\eta) = \langle 0 \rangle \implies \eta$ is injective, unless $0 \in U$.

Example 6.2. TODO: Let R be any ring and $U = \{0, 1\}$. Then

$$U^{-1}R = \left\{ \frac{r}{0} \mid r \in R \right\} \cup \left\{ \frac{r}{1} \mid r \in R \right\}$$

Let $\frac{r}{1} \stackrel{?}{=} \frac{0}{1} \iff t(r - 0) = 0 \iff tr = 0$ for some $t \in U$. We can take $t = 0$ and make this always true (same argument $\frac{r}{0} = \frac{0}{1}$). Hence $U^{-1}R$ is the zero ring.

Lemma 6.3. Let $U \subseteq R$ be multiplicatively closed. If U contains a nilpotent then $U^{-1}R$ is the zero ring.

Definition 6.4. If $P \subseteq R$ is a prime ideal then $U = R \setminus P$ is multiplicatively closed. Here we write $U^{-1}R = R_P$ and say R **localized at** P .

Lemma 6.5.

$$PR_P = \left\{ \frac{r}{u} \mid r \in P, u \in R \setminus P \right\} \subseteq R_P$$

is a maximal ideal.

Proof. Let $PR_P \subsetneq I \subseteq R_P$. Then for $r \in I$ $\frac{r}{u}$ with $r \notin P \implies \frac{r}{u}$ with $r, u \in R \setminus P \implies \frac{u}{r} \in R_P \implies \frac{r}{u}$ is a unit in $R_P \implies I = R_P$. **TODO:** $\square$

Proposition 6.6. R_P is a local ring (i.e., PR_P is the unique maximal ideal in R_P).

Proof. If $I \subseteq R_P$ is a maximal ideal, then $I \cap R \setminus P \neq \emptyset$, that is I contains a unit, and hence $I = R_P$. $\square$

Example 6.3. Let $f \in R$, and $U = \{1, f, f^2, \dots\}$, then

$$U^{-1}R = R \left[\frac{1}{f} \right] = \left\{ \frac{r}{u} \mid r \in R, u \in U \right\} = \left\{ \frac{r}{f^k} \mid r \in R, k \in \mathbb{Z}_{\geq 0} \right\}$$

that is $U^{-1}R$ is the ring of polynomials in $1/f$ with coefficients in R .

Lemma 6.7.

$$R \left[\frac{1}{f} \right] \cong \frac{R[x]}{\langle xf - 1 \rangle}$$

Proof. $R[x] \rightarrow R\left[\frac{1}{f}\right]$, $x \mapsto \frac{1}{f}$ and $1 \mapsto \frac{1}{f}$. This is surjective because for every $rx^k \in U^{-1}R$ we have $rx^k \mapsto \frac{r}{f^k}$. To show injectivity we must show that $\ker(\varphi) \supseteq \langle xf - 1 \rangle$ is trivial **TODO:** $\square$

Theorem 6.8. *There is an inclusion preserving bijection*

$$\begin{aligned} \{ \text{prime ideals of } U^{-1}R \} &\xleftrightarrow{1:1} \{ \text{prime ideals } P \subseteq R \text{ such that } P \cap U = \emptyset \} \\ P &\mapsto \eta(P) \end{aligned}$$

Let $Q \subseteq R$ be a prime ideal such that $Q \cap U \neq \emptyset$. Consider $\eta(Q) \cdot U^{-1}R = \langle \eta(Q) \rangle$. Let $r \in Q \cap U$, $\eta(r) \in \eta(Q) \subseteq \eta(Q) \cdot U^{-1}R$. Since $r \in U$, $\eta(r)$ is a unit in $U^{-1}R$, which implies

$$\eta(Q) \cdot U^{-1}R = U^{-1}R$$

i.e. it's not prime. **TODO:**

Theorem 6.9.

$$\{ \text{Spec}(U^{-1}R) \} \xleftrightarrow{1:1} \{ P \in \text{Spec}(R) \mid P \cap U = \emptyset \} \subseteq \text{Spec}(R)$$

$$P \mapsto \eta^{-1}(P)$$

Definition 6.10. Let R be a ring, then a basic open subset of $\text{Spec}(R)$ is of the form $D(f) = \{P \in \text{Spec}(R) \mid f \notin P\}$ for $f \in R$.

And,

$$D(f) = \text{Spec}(R) \setminus V(I) = \text{Spec}(R) \setminus V(\langle f \rangle)$$

The intersection of basic open sets $D(f) \cap D(g)$ is

$$D(f) \cap D(g) = \{P \in \text{Spec}(R) \mid f \notin P, g \notin P\} = \{P \in \text{Spec}(R) \mid fg \notin P\} = D(fg)$$

Remark 6.11. We know that there is a bijection,

$$\text{Spec}\left(R\left[\frac{1}{f}\right]\right) \leftrightarrow \{P \in \text{Spec}(R) \mid \{1, f, f^2, \dots\} \cap P \neq \emptyset\} = \{P \in \text{Spec}(R) \mid f \notin P\} = D(f)$$

Hence there is a bijection,

$$\text{Spec}\left(R\left[\frac{1}{f}\right]\right) \leftrightarrow D(f)$$

Theorem 6.12. Let $f \in R$ and $\eta : R \rightarrow R\left[\frac{1}{f}\right]$ be a localization map. The induced map

$$\text{Spec}\left(R\left[\frac{1}{f}\right]\right) \xrightarrow{\eta^\#} \text{Spec}(R)$$

is a homeomorphism onto $D(f)$.

Definition 6.13. The nilradical of R is the set of nilpotents

$$\mathfrak{N}(R) = \{f \in R \mid f^k = 0, \text{ for some } k\} = \sqrt{\langle 0 \rangle}$$

Proposition 6.14. $\text{Spec}(\mathbf{R}) \cong \text{Spec}(\mathbf{R}/\mathfrak{N}(\mathbf{R}))$ induced by the quotient map

$$\mathbf{R} \xrightarrow{\pi} \mathbf{R}/\mathfrak{N}(\mathbf{R})$$

such that

$$\text{Spec}(\mathbf{R}/\mathfrak{N}(\mathbf{R})) \xrightarrow{\pi^\#} \text{Spec}(\mathbf{R})$$

Proposition 6.15. $\mathfrak{N}(\mathbf{R}) = \bigcap_{\mathbf{P} \in \text{Spec}(\mathbf{R})} \mathbf{P}$

Proof. ($\implies$)

Let $f \in \mathfrak{N}(\mathbf{R})$ so $f^n = 0$. Pick any prime ideal $\mathbf{P} \subseteq \mathbf{R}$, $f^n = 0 \in \mathbf{P} \implies f \in \mathbf{P}$ since $\mathbf{P}$ is prime.

($\impliedby$)

We show given $f \notin \mathfrak{N}(\mathbf{R})$ there exists a prime $\mathbf{P} \subseteq \mathbf{R}$ with $f \notin \mathbf{P}$ (i.e. $D(f) \neq \emptyset$). □